Frequentist, Bayesian and Machine Learning Analysis of the Matter–Antimatter Asymmetry in $B^{\pm} \to K^{\pm}K^{+}K^{-}$ Decays at LHCb

Final project for the *Statistics and Data Analysis* course
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Giacomo Moretti and Marco Belotti
Università degli Studi di Milano-Bicocca

[TODO: Giacomo + Marco, write last. 3–5 sentences: dataset (LHCb Open Data, $B^{\pm} \to K^{\pm}K^{+}K^{-}$), the three methodological approaches compared (frequentist MLE/profile likelihood, Bayesian MCMC, ML-assisted selection), the headline result (A_{raw} estimate + consistency across methods), one sentence on what's new/interesting (e.g. goodness-of-fit model comparison, local Dalitz asymmetries). Don't give every number here – see §VIII guidance.]

I. INTRODUCTION

[TODO: Shared. Physics motivation: CP violation, matter-antimatter asymmetry, why three-body charmless B decays are interesting (Dalitz structure allows both global and local asymmetry studies). Cite the LHCb Open Data record and the reference paper (arXiv:2403.19921). State the research question (see `Docs/Proposta_progetto_LHCb_gruppo.tex`, already written) and preview the three-way methodological comparison.]

II. DATA

[TODO: Shared – this section can largely reuse material already written/verified: dataset description (record 4900, 2011 data, $\sqrt{s} = 7\,\text{TeV}$), branch schema, and the shared preselection. See `Docs/Proposta_progetto_LHCb_gruppo.tex` §4 and `lhcb_bayesian_coding_notes` memory for the exact preselection cut and the real-data mass distribution findings (peak ∼5284 MeV, background rise below 5150 MeV).]

A. Dataset and Preselection

[TODO: Describe the B2HHH ntuples, the shared preselection cut, and the H1-is-always-bachelor convention discovered while implementing the Dalitz analysis.]

B. Invariant Mass Reconstruction

[TODO: Kaon-mass-hypothesis four-vector sum, fit window choice [5150,5450] MeV and why (background rise below 5150 MeV).]

C. Data Exploration

[TODO: Mass spectrum plot (already have this, generated in `bayes` notebook), magnet up/down split, class balance for the ML labels.]

III. FREQUENTIST ANALYSIS

[TODO: Marco. Mass model $f(m) = N_s S(m) + N_b B(m)$, simultaneous fit parametrized on A, MLE via iminuit, profile likelihood $q(A)$, confidence interval, systematic uncertainties. See `Code_general/Frequentist/` on branch `freq` once populated.]

A. Mass Model and Fit Strategy

B. Profile Likelihood and Confidence Interval

C. Systematic Uncertainties

IV. BAYESIAN ANALYSIS

[TODO: Giacomo – this section is the most ready to draft for real, branch `bayes` is complete. Pull directly from `Code_general/Bayesian/bayesian_model.py` and `results/posterior_summary.json`: model/priors, emcee sampling + convergence diagnostics, posterior results ($A_{\text{raw}} = -0.0497$, 68% CI $[-0.058, -0.041]$), Savage-Dickey Bayes factor (with the bandwidth-sensitivity caveat), goodness-of-fit model comparison (Gaussian χ^2/ndof=2.36 → Crystal Ball fixed=4.50 → Crystal Ball free=2.00), magnet up/down cross-check, toy-MC coverage check, systematic variations, Dalitz local asymmetry.]

A. Model and Priors

B. MCMC Sampling and Convergence Diagnostics

C. Posterior Results and Credible Intervals

D. Bayes Factor: $A = 0$ vs A Free

E. Goodness-of-Fit and Signal Model Comparison

F. Systematic Checks: Mass Window, Background Model, Magnet Polarity

G. Toy-MC Coverage Validation

V. MACHINE LEARNING MODULE

[TODO: Shared – branch `mach_lear` is complete. Pull from `Code_general/Machine_Learning/ML_Analysis.ipynb` and `results/ml_summary.json`: weak-label construction from sidebands, feature set, model comparison (LogReg/Tree/RF/GBM/FFNN, AUC table), regularization study, bias-variance curve, calibration (+ isotonic recalibration result), uncertainty quantification (MC Dropout vs RF ensemble-variance, and why they disagree), hyperparameter/architecture search, A_{raw} vs threshold stability with bootstrap errors, K-means on the Dalitz plot.]

A. Weak Labels and Feature Set

B. Model Comparison

C. Regularization, Bias-Variance, and Calibration

D. Uncertainty Quantification

E. K-means on the Dalitz Plot

F. Stability of A_{raw} under ML Selection

VI. COMPARISON AND DISCUSSION

[TODO: Shared, written last once §III exists – the one section that genuinely can't be finished without Marco. Headline comparison table: A_{raw} (counting, frequentist, Bayesian, ML-assisted) with uncertainties, side by side. Comparison with the published A_{CP}(KKK) value from arXiv:2403.19921 and discussion of why $A_{\text{raw}} \neq A_{CP}$. Local Dalitz-plot asymmetry: fixed binning vs K-means, and whether the frequentist local significance study (if Marco does one) agrees.]

VII. LIMITATIONS AND FUTURE WORK

[TODO: Shared. This section can be drafted almost entirely NOW by condensing the already-written "Limiti noti" lists – don't re-derive, translate/tighten what's already there:

- From `bayes`: background assumed symmetric between charges (not validated independently of the magnet cross-check); Savage-Dickey Bayes factor's exact value is bandwidth-fragile (only the qualitative conclusion is robust); toy-MC coverage used a limited number of replicas and showed a mild ($\sim 1.3\sigma$) hint of undercoverage, not fully resolved; even the best signal model (Crystal Ball, floated shape) gives $\chi^2/\text{ndof} \approx 2$, not ≈ 1; Dalitz local-asymmetry binning is coarse.

- From `mach_lear`: severe class imbalance in the weak labels (1217 background-like vs 15119 signal-like); MC Dropout and RF ensemble-variance uncertainty estimates disagree by $\sim 18\times$ in absolute scale (only weakly correlated per-event, $\rho \approx 0.24$); hyperparameter/architecture search used a non-exhaustive grid; the A_{raw}-vs-threshold check is simple post-cut counting, not a full mass refit on the ML-selected sample.

- Cross-cutting: the mass window, background model, and signal-region definition were chosen from the Bayesian fit alone and still need to be reconciled with Marco's frequentist choices; no detector-level systematic (efficiency, production asymmetry) correction is applied anywhere – A_{raw} is explicitly NOT A_{CP}.

Future work: sWeight-based (not simple-cut) local Dalitz asymmetry; full mass refit after ML selection instead of counting; extend the toy-MC coverage study with more replicas.]

VIII. CONCLUSIONS

[TODO: Shared. Short – synthesize ONE key message per method (does not need to restate every number from the Results sections), following the reference report's style: state what was learned methodologically (e.g. "a minimal Gaussian+exponential model is insufficient, a Crystal Ball with floated shape parameters improves the fit without moving A_{raw}"), not just what was measured.]

IX. AUTHOR CONTRIBUTIONS

Marco Belotti led the frequentist analysis (Sec. III): completion of the baseline notebook, mass model, and profile-likelihood confidence interval. Giacomo Moretti led the Bayesian analysis (Sec. IV): model definition, MCMC sampling, convergence diagnostics, Bayes factor, and the associated systematic/goodness-of-fit studies. The machine learning module (Sec. V), the phase-space/toy-MC validation work, the Dalitz-plot analyses, and the final cross-method comparison (Sec. VI) were carried out jointly. Both authors contributed to the manuscript.

X. CODE AND DATA AVAILABILITY

All code is written in Python ≥3.12, run in a dedicated `conda` environment (`cern`). Key package versions: `uproot 5.7`, `iminuit 2.32` **[TODO: Marco confirm]**, `emcee 3.1.6`, `corner 2.3.0`, `scikit-learn 1.9`, `tensorflow 2.21.0`, `keras 3.15.1`, `scipy 1.18`, `numpy 2.4`, `pandas 3.0`.

The dataset is the CERN Open Data record 4900 ("Matter Antimatter Differences – B meson decays to three hadrons"). Analysis code is organized per branch in the project git repository: `Code_general/Frequentist/`, `Code_general/Bayesian/`, `Code_general/Machine_Learning/`.

Appendix A: Full results tables

[TODO: Shared. Mirror the reference report's appendix style: every tested variant in one table, chosen/final result highlighted. Needs (a) Bayesian systematic-variation table (5 rows: nominal + 4 variants, already in `results/posterior_summary.json` on `bayes`), (b) ML model-comparison table (all 6 classifiers x AUC/Brier, already in `results/ml_summary.json`), (c) ML hyperparameter/architecture search table (RF/GBM grid + FFNN architecture search top results), (d) frequentist fit results once Marco has them.]